BATCH4_RTA_GEOM

PREREG: BATCH4_RTA_GEOM (C15 closeout; Map-C16, C17, C18)

Written 2026-10-24 at about 7:10 PM CDT, before any BATCH4 test. Owner: Arvin Hampton. New files only. CORE_FREEZE. All maps are **Category B**; a pass is a survived prediction, not a bridge. No quadratic-form → O_res maps. No Π, grading or T3-orbit carrier function is invented. order20 and the paper stream are not touched. Runs: nice, OMP_NUM_THREADS=1, ulimit -v 3000000. 67/68 are not targets.

Written residual objects (paths in NOTES): - Core $x_j = 21+9j$ ($j = 0..538$). - Fibres $\beta\sharp(x_j) = j \bmod 8$. - Towers τ: blocks of $\Omega = \{0..4879\}$, the first 20 of size 21 and the rest of size 20. - $f(i) = \tau(x_i)$, $\delta f(i) = f(i+1) - f(i)$ on path edges e_i, $i = 0..537$. - Product complex $X_\times = K_9 \times P_{B'}$ with $\alpha(a,b) = \mathrm{sgn}(a-b)$ and the mixed cochain $\alpha \otimes \delta f$ on squares $S(a,b;i)$ (Residual_Product_Complex.md).

C15 closeout (prediction unchanged from BATCH3 PREREG, sha256 b57478f7…21b2)

$k := \#\{\text{companions } c \in 0..104 : \max \text{ over the 216 points of } |A2_corner[:,c]| > 1e{-}10\}$, read from order18/data/a2_arrays.npz, array CO (A2_corner, RG4). Gap guard: the result is VOID if any column max lies in $[1e{-}13, 1e{-}7]$. PASS iff $8 \mid k$.

Map-C16: β♯ equidistribution of tower changes (RG1′-type sector split → fibres)

Source: the P-odd/P-even split is balanced by the RG1′ reflections (order16). Target: the edges with $\delta f(i) \neq 0$, classed by β♯ of the start index, $i \bmod 8$. Prediction (NEW): the 8 class counts satisfy $\max - \min \leq 1$. Gate: $\Sigma\delta f = 241$ (written). If the gate fails → VOID.

Map-C17: palindromic tower occupancy (endpoint reflection ρ → core reversal)

Source: the RG1′ endpoint reflection. Target: $n_t = \#\{j : \tau(x_j) = t\}$ over the touched towers, in tower order. Prediction (NEW): the sequence (n_t) is a palindrome. Gate: $\Sigma_t C(n_t, 2) = 353$ and 242 touched towers (both written). If the gate fails → VOID.

Map-C18: permanent-class fibre pairing has the reversal symmetry

Source: P-even corner A2c (order18; P-even = invariant under the reflection P). Target: v_r = ⟨α⊗δf, Σ_{i ≡ r (8)} S(0,8;i)⟩ = α(0,8)·Σ_{i≡r} δf(i), r = 0..7. The core reversal sends edge i ↦ 537 − i, which induces r ↦ (1 − r) mod 8. Prediction (NEW): v_r = v_{(1−r) mod 8} for all r (“P-even” on the fibres). PASS iff it holds exactly. # RESULTS: BATCH4_RTA_GEOM PREREG.md sha256 d4f727655004cf599d2a0c2564109913c02f8e0a30ab9dbb8b3474c0aa189475, hashed before tests. Script scripts/b4.py (hash in scripts/SHA256SUMS); data data/b4.json; log logs/b4.log. Under 1 s.

Map	Prediction	Measured	Verdict	Cat
C15 closeout	8 | k, k = #companions with A2c ≠ 0	k = 105 (every column nonzero; min column max 6.14; no gap hit)	**FAIL** (105 = 8·13 + 1)	B (float data from order18 CO = corner(), RG4)
C16	tower-change edges are equidistributed over i mod 8 (max − min ≤ 1)	counts [27, 28, 40, 26, 27, 27, 39, 27]; gate Σδf = 241 OK	**FAIL**	B (exact integers)
C17	tower occupancy (n_t) is a palindrome	gates OK (242 towers, 353 pairs); head 3,2,2,3,… vs tail …,2,2,1	**FAIL**	B (exact)
C18	v_r = v_{1−r} on fibres (P-even reversal symmetry)	v = −[27, 28, 40, 26, 27, 27, 39, 27]; v_0 = −27 vs v_1 = −28	**FAIL**	B (exact)
C6, C10, C11, E1–E3	—	still no written Π, ±1 grading or path predicate	**OPEN**	B

No bridge passed. No bridge claim.

Post-hoc (not preregistered, no verdict change): - The C16/C18 peaks at classes 2 and 6 come from the arithmetic of 9j against tower widths 21/20. Mod 8 these align with period 4. This is a carrier-internal artifact, not modular structure. - C15: A2c is nonzero on every one of the 105 companions, so A2c has full support. This agrees with order18/19 (“P-even and nonzero”). # NOTES: BATCH4_RTA_GEOM ## Sources re-read - Product complex, α = sgn(a−b), f = τ(x_i), δf, α⊗δf on squares: /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/Residual_Product_Complex.md (copies also in oct17-master/docs/…, reports/self-audit/repo/docs/…, reports/bridges/*git_v8_unpushed_clone/docs/…). - Permanent-class survival (optional*

lock): …/P_plus_Permanent_Class_Survival_Optional_Lock_2026-08-08.md (same trees). It is a cohomology class on X×. No path evaluation is defined (consistent with the predicate-sources report). - Core, β♯, τ, 242/241/353: reports/crypto/RPP_spec_v1/RESIDUAL_T3_predicate_sources.md §2.2–2.5. The probe definitions are cited there (Object539_NonCircular_Construction, Discrete_Patterns_Residual_Carrier). - Path-predicate rule (no carrier functions on T3 orbits): same file §3; respected. - C15 data: order18/data/a2_arrays.npz, array CO, written by order18/scripts (line 66: CO[ix,j] = corner(al,be,x)). order19 was read only.

Remarks

- C16–C18 are the fibre-faithful (j mod 8) maps. C4 is not reopened.
- The C18 pairing over all fibres totals −241 = α(0,8)·Σδf, consistent with the written Σδf.
- **Sharpest open next step:** all cheap β♯/τ/permanent-class symmetry predictions so far fail, because the carrier's 9j vs 21/20-width arithmetic has period-4 structure in j mod 8 that the modular side does not see.
  - The remaining well-posed lever is a *cohomological* rather than numerical map: predict that the permanent class [α⊗δf] restricted to the β♯-fibre subcomplexes K_9 × P_r (r = 0..7) is non-trivial for exactly the r singled out by a written modular datum. This needs the written survival criterion from P_plus_Permanent_Class_Survival (thin form complex), computed per fibre.
  - Otherwise, the Resonant targets stay blocked on a written Π / grading.
- order20 and the paper stream were not touched. CORE_FREEZE; new files only; no push, no sends.